# AIRFOIL SELECTION FOR TWO RC AIRCRAFT MISSIONS


**EXECUTIVE SUMMARY**

This report compares candidate airfoils for two subsonic radio-controlled (RC) aircraft: **Aircraft A**, designed to maximize payload capacity while maintaining stable low-speed flight, and **Aircraft B**, designed for high aerobatic manoeuvrability during turns, rolls, and inverted flight. Three candidates are shortlisted for each mission based on geometric characteristics, lift production, drag efficiency, stall behavior, and structural suitability. Preliminary selections recommend the high-camber **Selig S1223** for Aircraft A and the symmetrical **NACA 0012** for Aircraft B.


## 1. Assumptions and Comparison Conditions

Both aircraft are assumed to be small, fixed-wing, subsonic RC vehicles. Because specific gross mass, wing area, and cruise velocity profiles were not pre-defined, direct aircraft-level performance predictions are unconstrained. To establish a rigorous common baseline for section-level comparison, standard operating conditions are defined at a reference chord $c$ = 0.20 m and flight velocity $V$ = 15 m/s in standard air (kinematic viscosity $\nu = 1.5 \times 10^{-5}$ m²/s). This yields a nominal Reynolds number:

$$\mathbf{Re = V \cdot c / \nu \approx 200{,}000}$$

## 2. Theoretical Background

**2.1 Airfoil Geometry:** Camber defines the asymmetric curvature of the mean line relative to the chord line. Cambered sections generate positive lift at zero angle of attack (α = 0°), whereas symmetrical sections produce zero net lift at α = 0°. Symmetrical profiles provide identical performance inverted, making them essential for precision aerobatics. Section thickness dictates internal structural volume for spars while influencing pressure gradients.

**2.2 Lift, Drag, and Efficiency:** Aerodynamic lift ($L$) and drag ($D$) forces are governed by $L = \frac{1}{2} \rho V^2 S C_L$ and $D = \frac{1}{2} \rho V^2 S C_D$. Section efficiency is represented by the lift-to-drag ratio ($L/D = C_L/C_D$). High maximum lift coefficient ($C_{L,max}$) is critical for heavy payloads, whereas low zero-lift drag coefficient ($C_{D0}$) is crucial for high-speed maneuvering.

**2.3 Low Reynolds Number Dynamics & Stall:** Operating at Re ≈ 200,000 subjects profiles to boundary layer laminar separation bubbles, increasing drag and precipitating sudden stall if unmanaged. Gentle stall behavior is paramount for heavy payload recovery.

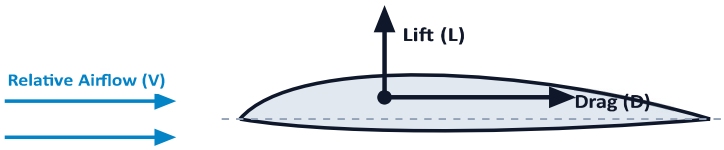



*Figure 1. Schematic aerodynamic force vectors acting on a cambered wing section relative to incoming airflow.*

## 3. Airfoil Shortlisting and Preliminary Selection

### 3.1 Aircraft A — Maximum Payload with Stable Flight

The primary design goal for Aircraft A is maximizing low-speed payload transport while maintaining passive pitching stability and forgiving stall dynamics. Three high-lift candidate airfoils were shortlisted:

- **Selig S1223:** Purpose-designed for low Reynolds number high-lift applications. Published wind-tunnel data by **Selig & Guglielmo (1997)** confirms an extraordinary maximum lift coefficient of $C_{L,max} \approx 2.2$ at Re = 200,000. Its deep undercamber generates immense lift, though it exhibits significant nose-down pitching moment ($C_m$) requiring robust tail trim.
- **NACA 4412:** A classic cambered profile featuring 4% maximum camber at 40% chord with 12% thickness. It offers moderate lift capabilities with well-behaved stall characteristics and lower pitching moment than high-undercamber sections.
- **Clark Y:** A historic profile with 3.4% camber and 11.7% thickness. Famous for its flat lower surface which simplifies airframe construction, it provides highly reliable low-speed operation and smooth stall behavior.

*Preliminary Selection for Mission A:* **Selig S1223**, selected due to its superior raw lift performance to maximize payload margin.

### 3.2 Aircraft B — Manoeuvrability and Aerobatics

Aircraft B demands symmetric force generation in upright and inverted attitudes, quick roll response, and predictable inverted stall behavior.

- **NACA 0012:** A standard 12% thick symmetrical profile. Provides identical lift curves in upright and inverted flight, zero pitching moment at zero angle of attack, and excellent control surface response.
- **NACA 0015:** A 15% thick symmetrical profile. The increased thickness provides structural depth for wing spars and slightly gentler stall characteristics, though with higher section drag.
- **NACA 2412:** A semi-symmetrical section with 2% camber. Offers higher upright efficiency but asymmetric inverted response.

*Preliminary Selection for Mission B:* **NACA 0012**, selected for its ideal symmetric aerobatic baseline.

## 4. Comparative Geometry Profiles

The six shortlisted candidate profiles illustrate distinct geometric approaches to balancing lift, drag, pitching moment, and structural spar thickness across low Reynolds number regimes.

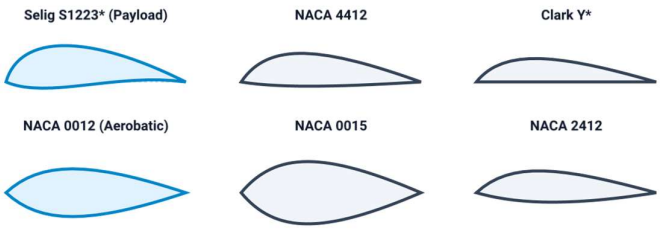



*Figure 2. Profile geometry comparison of shortlisted airfoils (*S1223 and Clark Y shown schematically).*

## 4. Comparative Evaluation Matrix (Continued)

Below is the systematic comparative summary for all six shortlisted candidate airfoils across both mission classes:

| Airfoil / Mission | Geometry | Primary Rationale | Main Technical Limitation |
|---|---|---|---|
| **Selig S1223** (Aircraft A) | Highly cambered undercamber profile | Exceptional low-Re lift ($C_{L,max} \approx 2.2$); maximizes payload payload weight. | High nose-down pitching moment ($C_m$) and deep undercamber construction effort. |
| **NACA 4412** (Aircraft A) | 4% camber, 12% thickness | Proven conventional cambered profile; balanced lift-to-drag and gentle stall. | Lower maximum lift ceiling compared to specialized undercamber profiles. |
| **Clark Y** (Aircraft A) | 3.4% camber, 11.7% thickness | Flat bottom simplifies wing construction; reliable low-speed stability. | Moderate peak lift capacity compared to high-lift S1223. |
| **NACA 0012** (Aircraft B) | 0% camber (Symmetric), 12% thick | Identical inverted performance; zero $C_m$; smooth roll rate control. | Lower upright cruise efficiency compared to cambered sections. |
| **NACA 0015** (Aircraft B) | 0% camber (Symmetric), 15% thick | Deep section allows stronger internal spars; softer stall margin. | Higher pressure drag at higher velocities. |
| **NACA 2412** (Aircraft B) | 2% camber, 12% thickness | Versatile compromise profile for general utility aircraft. | Asymmetric performance during inverted aerobatic maneuvers. |

## 5. Whole-Aircraft Integration & Mission Trade-Offs

Selecting an optimal airfoil profile is only one component of complete aircraft design. Wing loading ($W/S = m \cdot g / S$) and maximum lift coefficient jointly dictate the level flight stall speed:

> **Stall Speed Equation:** $V_s = \sqrt{( 2 \cdot W / ( \rho \cdot S \cdot C_{L,max} ) )}$

For **Aircraft A (Payload)**, deploying the Selig S1223 maximizes $C_{L,max}$, which directly depresses stall speed $V_s$ for a given total mass. This permits carrying heavier payloads within safe hand-launch or short-takeoff operational constraints. However, the strong nose-down pitching moment of S1223 requires larger horizontal tail area or negative tail trim, introducing trim drag that must be offset by the propulsion system.

For **Aircraft B (Aerobatic)**, deploying NACA 0012 provides symmetrical lift curves. Aircraft agility depends heavily on wing aspect ratio, roll moment of inertia, control surface sizing, structural spar stiffness, and thrust-to-weight ratio (**NASA Glenn, 2021**).

## 6. Dual-Mission Feasibility Analysis

A central system engineering question is whether a single, dual-purpose airframe can satisfy both mission profiles. While feasible, a dual-purpose design inherent requires performance trade-offs.

A compromise profile like the **NACA 2412** offers moderate camber for payload lift alongside benign handling, but lacks inverted symmetry. Alternatively, using **NACA 0012** on a dual-role airframe guarantees aerobatic capability, but demands higher speed or larger wing area to carry heavy payloads. If both missions demand maximum performance, two dedicated aircraft are recommended.




*Figure 3. Recommended preliminary engineering selection and validation workflow.*

## 7. Limitations and Conclusion

This evaluation provides a preliminary section-level analysis. Final section selection must be validated using precise flight speeds, wing planform geometries, structural weight build-ups, and experimental windtunnel or computational verification (**Virginia Tech, 2021**).

In conclusion, the **Selig S1223** is recommended for Aircraft A to maximize payload capacity, while the **NACA 0012** is recommended for Aircraft B as a benchmark aerobatic platform. **References**


1. Selig, M. S., & Guglielmo, J. J. (1997). High-Lift Low Reynolds Number Airfoil Design. *Journal of Aircraft*, 34(1), 72–79.
2. NASA Glenn Research Center (2021). *Inclination Effects on Drag & Drag Coefficient Analysis*. NASA Educational Aeronautics Series.
3. Virginia Tech AOE Department (2021). *Experiment 7: Low Reynolds Number Aerodynamics of the Clark Y Airfoil*. AOE 3054 Instructional Manual.